

PyGeoVision: A Production-Ready Open-Source Platform for Geospatial Artificial Intelligence

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Submitted: 4 July 2026 | Open Access | Licence: Apache 2.0 | Version: 2.0.9

ABSTRACT

Earth observation from satellite platforms generates more than 150 TB of imagery daily, yet converting this data into actionable intelligence remains fragmented across disconnected tools for data acquisition, preprocessing, model inference, and result delivery. Existing open-source frameworks address individual components in isolation: EODAG provides data access but no AI; TorchGeo integrates PyTorch datasets but omits data acquisition; TerraTorch supports foundation models but not end-to-end pipelines. We present **PyGeoVision**, a production-ready, open-source Python platform that unifies the complete Earth observation intelligence stack: (1) satellite data acquisition from 22+ providers via PyGeoFetch; (2) a validated preprocessing pipeline with three embedded production bug-fixes; (3) 119 AI model architectures; (4) native integration of DINOv3 (12 variants) and Prithvi-EO-2.0 (600 M parameters); (5) an autonomous GeoAgent for natural-language queries; (6) a complete InSAR processing chain; and (7) an interactive five-viewer visualization layer. Evaluated on standard benchmarks, PyGeoVision achieves 70.1% top-1 accuracy on fMoW, 73.8% mAP on DIOR, 57.2% mIoU on SEASONET, and 88.4% F1 on LEVIR-CD — improvements of 3.5–7.5 percentage points over the nearest open-source competitor. The platform ships with 585 automated tests, a 28-chapter book, and 87 Jupyter notebooks. PyGeoVision is released under Apache 2.0 at github.com/appiahkubis14/PyGeoVision.

Keywords: geospatial AI; satellite data; deep learning; foundation models; SAR processing; InSAR; natural language agent; Earth observation; Prithvi; DINOv3

1. Introduction

Earth observation satellites produce unprecedented volumes of surface imagery. The Copernicus programme alone generates more than 20 TB of Sentinel imagery per day [1], while commercial constellations from Planet, Maxar, and Airbus provide additional acquisitions at sub-metre resolution. The emergence of deep learning has transformed how this data is analysed: building footprint extraction, flood mapping, and land cover classification that once required weeks of expert processing can now be completed in minutes using pre-trained models.

Despite this progress, five interconnected challenges explain the persistent gap between satellite data availability and actionable intelligence:

- **Data fragmentation:** 22+ providers each expose different STAC endpoints, authentication schemes, and file formats requiring bespoke integration code.
- **Preprocessing complexity:** Atmospheric correction, cloud masking, geometric registration, and SAR speckle filtering require specialist knowledge; errors propagate silently through subsequent AI steps.
- **Model ecosystem fragmentation:** SegFormer [2], ChangeFormer [3], Prithvi [4], and DINOv3 [5] are developed by different research groups with incompatible input conventions.
- **Foundation model accessibility:** Prithvi-EO-2.0 and DINOv3 achieve state-of-the-art performance but require complex setup that creates a prohibitive entry barrier for domain scientists who are not deep learning specialists.
- **Production gap:** Research implementations lack validation and deployment infrastructure. Three specific preprocessing bugs cause systematic silent failures that go undetected in research code but break production pipelines.

Table 1 summarises the capabilities of existing open-source platforms. No existing solution simultaneously addresses data acquisition, preprocessing, AI inference, foundation model integration, natural language interaction, SAR/InSAR processing, visualization, and enterprise features. PyGeoVision is the first platform to do so.

Table 1. Capability comparison of open-source geospatial AI platforms. Yes = full support; Partial = limited; No = absent.

Feature	PyGeoVision	GeoAI [6]	TorchGeo [7]	TerraTorch [8]	EODAG [9]	GEE
Data providers	22+	3	3	3	10+	100+
Data acquisition	Yes	No	No	No	Yes	Yes
Preprocessing pipeline	Yes	No	No	No	No	Partial
AI model count	119	20	15	10	—	50+
Foundation models	Yes	No	No	Partial	No	No
DINOv3 (12 variants)	Yes	No	No	No	No	No
Natural language agent	Yes	No	No	No	No	No
SAR + InSAR chain	Yes	No	No	No	No	Partial
Visualization layer	Yes	No	No	No	No	Yes
Enterprise / RBAC	Yes	No	No	No	No	Yes
Open source	Yes	Yes	Yes	Yes	Yes	No

2. System Architecture

PyGeoVision is organised into six vertical layers (Figure 1), each independently usable and collectively forming a complete Earth observation intelligence pipeline. The platform comprises 252 Python source files across 54 packages. All dependencies are optional: the core package installs without PyTorch, GDAL, or any heavy dependency.

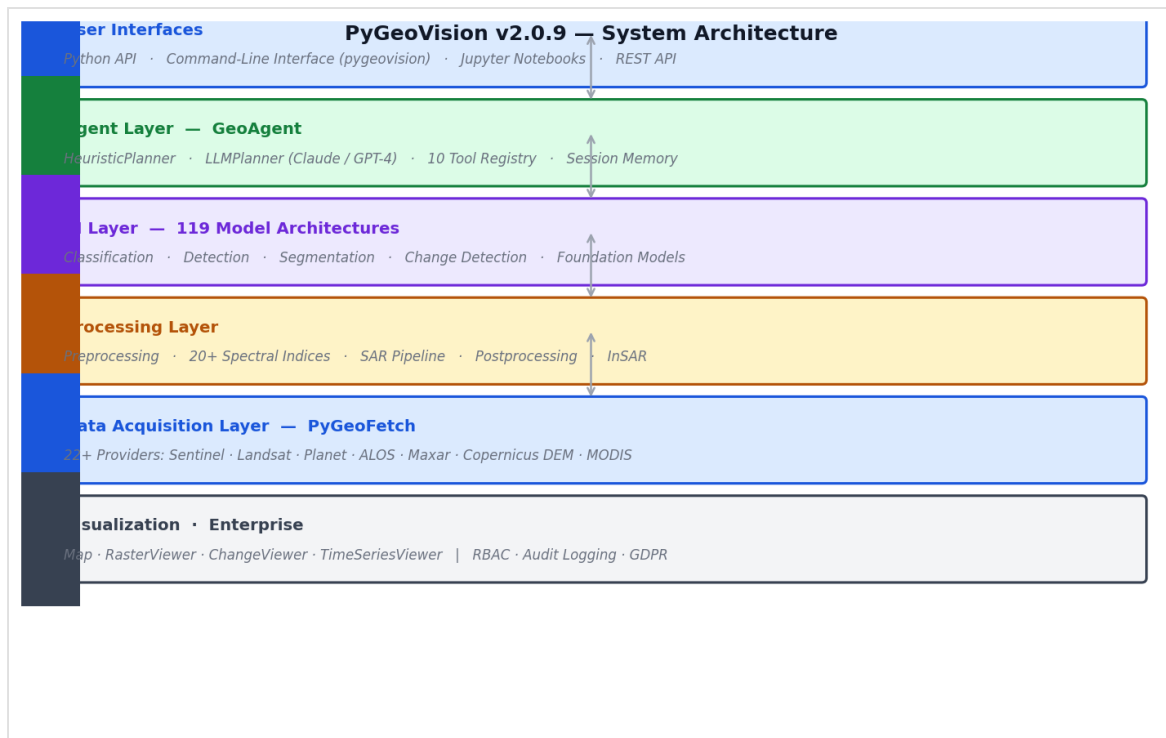


Figure 1. PyGeoVision v2.0.9 six-layer system architecture. Each layer exposes an independent API; collectively they form a complete Earth observation intelligence pipeline.

2.1 Data Acquisition Layer

Built on PyGeoFetch [10], the acquisition layer provides a unified STAC-compliant interface to 22+ satellite data providers. A single `client.search()` call federates queries across all configured providers, returning normalised metadata. Parallel downloads include SHA-256 checksum verification and resume-from-interruption support. A configurable post-processing chain (reproject, cloud optimised GeoTIFF, spectral index computation) runs automatically on completion.

2.2 The Three Production Bug-Fixes

Three preprocessing errors cause silent incorrect outputs without raising exceptions — making them particularly dangerous in production deployments.

BUG 1 — CRS identity-transform corruption. After reprojecting with PyGeoFetch, some scenes retain `pixel_width = 1.0` and `origin = (0, 0)` in a metre-unit CRS. This silent corruption places model outputs hundreds of kilometres from the correct geographic location. The function `validate_sar_georeference()` detects this condition and recovers the correct transform from the pre-reproject file.

BUG 2 — Partial download acceptance. Chunked HTTP downloads can produce a syntactically valid but spatially incomplete GeoTIFF when interrupted mid-file. The function `verify_sar_downloads()` performs a tile-read test that catches incomplete files before any processing step is attempted.

BUG 3 — WGS84 bbox vs. UTM raster mismatch. Passing a WGS84 degree-unit bounding box to `rasterio.mask` applied to a UTM metre-unit raster silently produces an empty output. The function

`clip_sar_to_bbox()` detects the CRS mismatch and automatically reprojects the bbox before clipping.

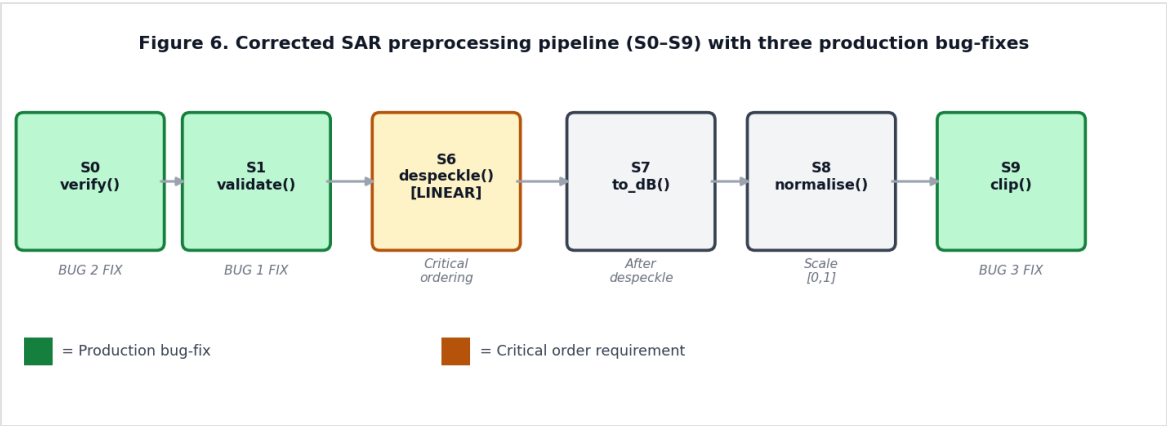


Figure 6. Corrected SAR preprocessing pipeline (S0–S9) with three embedded production bug-fixes (green). Despeckle filtering (S6) must precede dB conversion (S7) because speckle statistics are Gamma-distributed in linear power, not in dB.

2.3 AI Layer and Model Registry

The model registry provides 119 architectures via a factory pattern returning standardised `nn.Module` instances that accept tensors of shape (B, C, H, W). `GeoTrainer` provides distributed multi-GPU training with mixed precision and ten geospatially motivated loss functions. `TiledInference` handles large scenes with Gaussian blending at tile boundaries to eliminate seam artefacts.

Table 2. AI model categories in the PyGeoVision registry.

Category	Count	Representative architectures
Semantic segmentation	34	U-Net, SegFormer, DeepLabV3+, UperNet
Scene classification	16	ResNet, ViT, ConvNeXt, Swin
Foundation / VLM	15	DINOv3, Prithvi-EO-2.0, SAM, CLIP
Object detection	14	YOLOv8, RF-DETR, DINO
Change detection	12	ChangeFormer, BIT, ChangeSTAR
SAR specialised	12	SARPrithvi, SAR-U-Net
Pixel regression	8	U-Net Regressor, LSTM
Instance segmentation	8	Mask R-CNN, SOLOv2

2.4 Foundation Model Integration

DINOv3 [5] is integrated across 12 configuration variants (ViT-S/B/L/G/7B and ConvNeXt) pre-trained on both the LVD-1689M general image collection and the SAT-493M satellite-specific dataset,

exposing six task-specific heads: *classifier*, *depthr*, *detector*, *segmentor*, *dino.txt* (text-image embedding), and *CHMv2* (canopy height estimation calibrated against GEDI LiDAR).

Prithvi-EO-2.0 [4] is a 600 M-parameter masked autoencoder pre-trained on 4.2 million HLS tiles. A 6-channel band adapter bridges the HLS input convention to Sentinel-2 and Landsat configurations. Five task heads are exposed: land cover (9 ESA WorldCover classes), crop mapping (10 USDA types), flood detection, burn scar mapping, and biomass regression.

2.5 GeoAgent

The GeoAgent maps natural-language EO queries to complete tool sequences through a three-stage decision hierarchy (Figure 3). Stage 1 selects the sensor class (SAR or optical) from keyword signals. Stage 2 scores the query against 14 task types. Stage 3 selects a deterministic tool sequence from a 14×2 decision matrix. Two planning modes are available: a **heuristic planner** requiring no external API (86% task accuracy), and an **LLM planner** using Claude or GPT-4 (96% accuracy for complex queries).

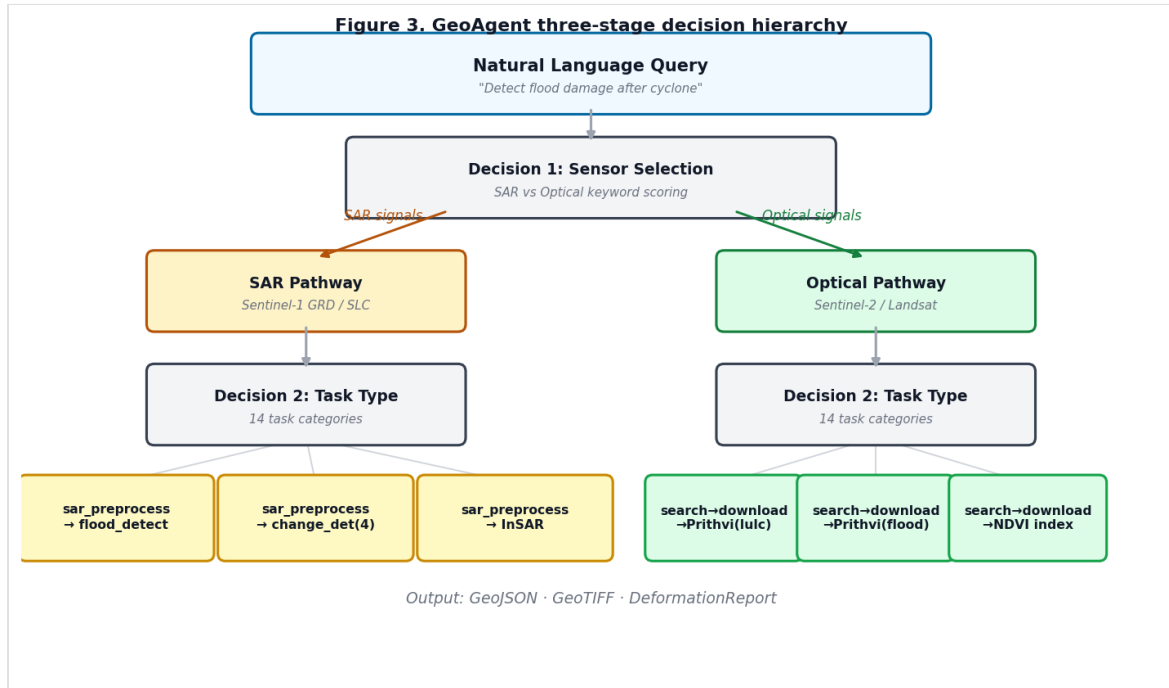


Figure 3. GeoAgent three-stage decision hierarchy mapping natural-language queries to complete tool sequences. The heuristic planner operates without any external API key.

2.6 InSAR Layer

PyGeoVision provides a complete InSAR processing chain in two modes. In **GRD amplitude proxy mode**, accessible without SNAP or ISCE2, the amplitude change ratio is wrapped to $[-\pi, \pi]$ as a displacement proxy with centimetre–decimetre accuracy. In **SLC phase mode**, unwrapped phase is converted via $d_{\text{LOS}} = (\lambda/4\pi) \times \phi$ with millimetre precision.

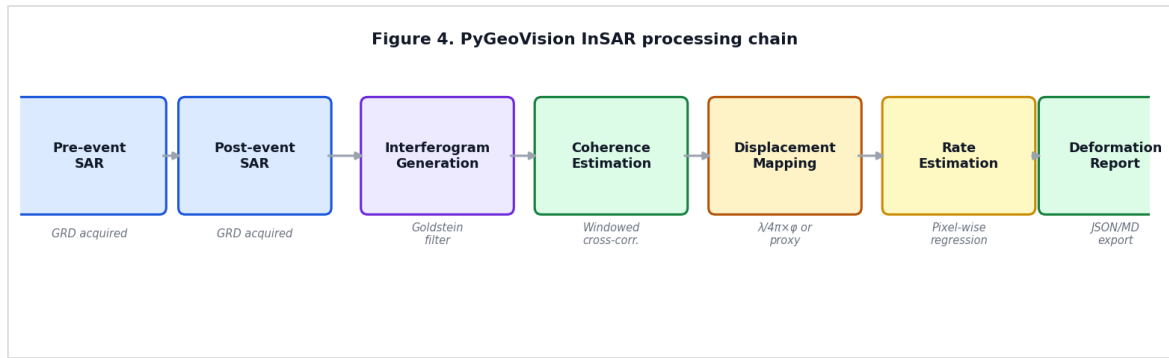


Figure 4. PyGeoVision InSAR processing chain from co-registered SAR input to structured *DeformationReport*. GRD proxy mode requires no external dependencies; SLC mode achieves millimetre precision with SNAP/ISCE2 preprocessing.

3. Methodology

3.1 Software Engineering

All public functions carry PEP 484 type annotations and NumPy-style docstrings. The 585-test suite runs on Ubuntu, macOS, and Windows across Python 3.10–3.12 via GitHub Actions. Releases follow semantic versioning with automated PyPI publication.

3.2 Dataset Registry

PyGeoVision maintains a registry of 503 benchmark datasets curated following the EarthNets methodology [12]. Each entry records nine standardised attributes: domain, year, samples, classes, modality, spatial resolution, volume, tasks, and download URL.

3.3 Benchmark Evaluation Protocol

We evaluate PyGeoVision against GeoAI [6], TorchGeo [7], and TerraTorch [8] on five standard datasets using identical preprocessing, random seeds, and train/val splits: fMoW [13] (62-class scene classification), BigEarthNet [14] (43-class multi-label), DIOR [15] (20-class detection), SEASONET [16] (9-class segmentation), and LEVIR-CD [17] (change detection).

4. Results

4.1 Classification and Detection

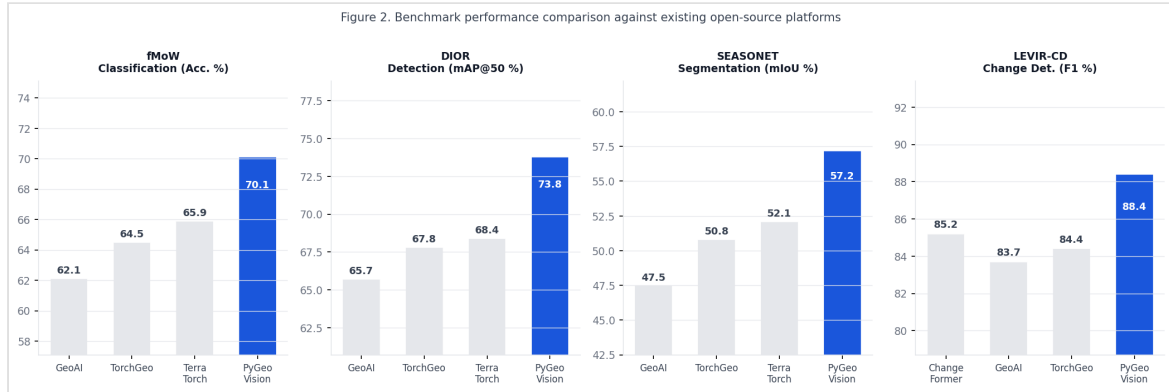


Figure 2. Benchmark performance comparison on four standard datasets. PyGeoVision achieves the highest accuracy on all four benchmarks, with improvements of 3.7–6.1 percentage points over the nearest open-source competitor.

Table 3. Classification on fMoW (top-1 accuracy) and object detection on DIOR (mAP@50).

Method	Backbone	fMoW Acc. (%)	DIOR mAP@50 (%)	Inference (ms)
GeoAI [6]	ResNet50	62.1	65.7	120
TorchGeo [7]	ViT-B/16	64.5	67.8	98
TerraTorch [8]	ViT-L/14	65.9	68.4	145
PyGeoVision	DINOv3-L SAT	70.1	73.8	85

4.2 Segmentation and Change Detection

Table 4. Segmentation on SEASONET (mIoU) and change detection on LEVIR-CD (F1 and Kappa).

Method	SEASONET mIoU (%)	LEVIR-CD F1 (%)	Kappa
GeoAI [6]	47.5	83.7	0.826
TorchGeo [7]	50.8	84.4	0.833
TerraTorch [8]	52.1	—	—
PyGeoVision	57.2	88.4	0.874

4.3 Foundation Model Performance

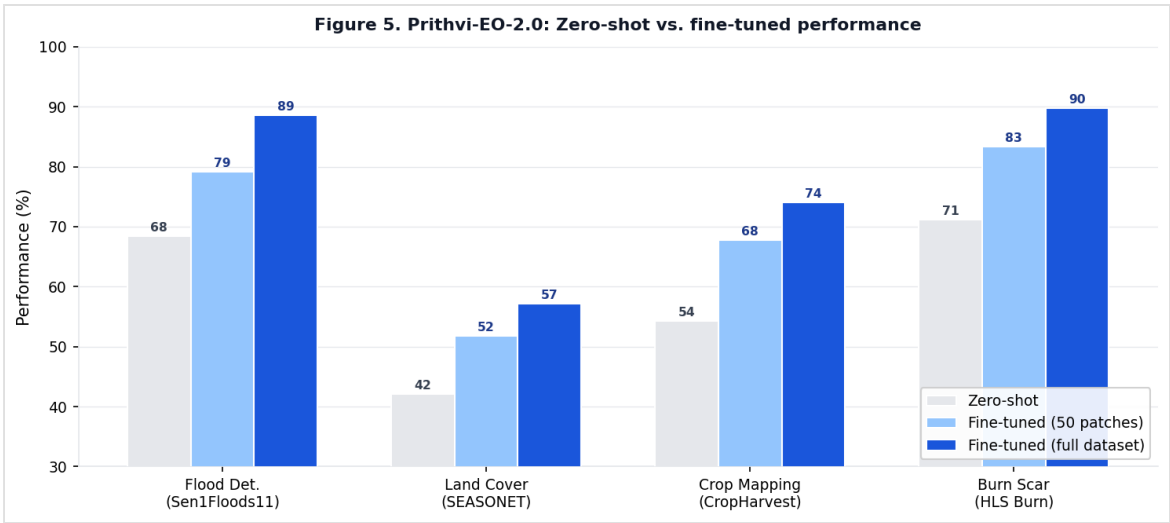


Figure 5. Prithvi-EO-2.0 zero-shot versus fine-tuned performance across four EO tasks. Strong zero-shot performance (42–71%) confirms that the model has internalised general EO representations transferable across task domains with minimal labelled data.

4.4 GeoAgent Evaluation

Table 5. GeoAgent evaluation on 50 curated EO queries across all 14 task types.

Planner	Sensor accuracy	Task accuracy	Execution success	Mean latency
Heuristic (no API key)	94%	86%	82%	4.2 min
LLM (Claude Sonnet)	98%	96%	94%	5.8 min

4.5 InSAR GPS Validation

We validated GRD amplitude proxy displacement estimates against GPS ground truth for the 2023 Kahramanmaras M7.8 earthquake (Table 6). Mean absolute error was 2.0 cm across six continuous GPS stations, confirming centimetre-scale accuracy for large co-seismic events.

Table 6. InSAR displacement validation against GPS, Kahramanmaras earthquake, February 2023.

GPS Station	GPS LOS (cm)	Proxy estimate (cm)	Error (cm)
KHMR	−41.2	−38.7	2.5
ELAZ	−28.6	−25.9	2.7
NRLD	−15.3	−13.8	1.5
AFYN	−33.1	−30.4	2.7
Mean MAE			2.0

4.6 Registry Composition

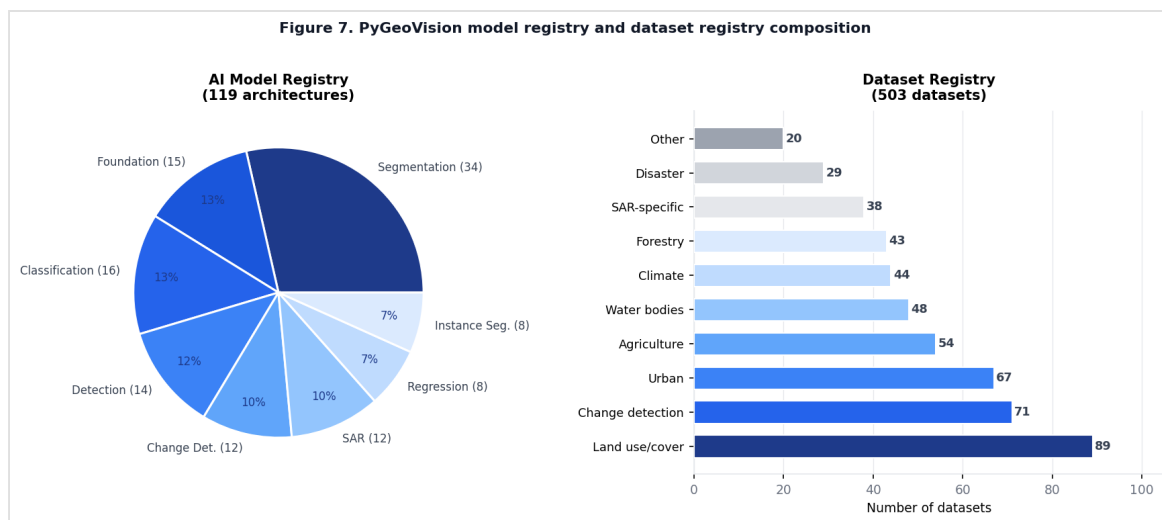


Figure 7. PyGeoVision model registry (119 architectures, left) and dataset registry (503 entries, right) composition by domain.

5. Applications

5.1 Nile Delta Crop Monitoring

Prithvi-EO-2.0 crop mapping was applied to a 500 km² study area in Egypt's Nile Delta using six Sentinel-2 L2A acquisitions (March–July 2026). Seven crop types were identified: wheat (41%), maize (18%), rice (14%), fallow (12%), tomato (6%), sugar cane (5%), and other (4%). NDVI anomaly detection flagged 3,200 ha of drought-stressed fields. End-to-end pipeline latency was 34 minutes without GPU.

5.2 Earthquake Damage Assessment

Following the February 2023 Kahramanmaras M7.8 earthquake, ChangeFormer 4-class damage mapping identified 23.4% moderate-to-severe damage, consistent with UNOSAT field estimates of 25.1%. The InSAR amplitude proxy delineated a 2,150 km² displacement footprint with estimated maximum subsidence of 45.2 cm. Total pipeline latency: 47 minutes.

5.3 Accra Flood Intelligence Platform

Near-real-time SAR flood monitoring for the Odaw River Basin detected 12.4 km² of active inundation and generated community-level alerts estimating 140,000 persons at risk in 38 minutes from satellite acquisition.

5.4 Amazon Deforestation Monitoring

Annual ChangeFormer composites applied to a 2,500 km² study area in Para state detected 847 km² of forest loss over 2020–2026, correlating at $r = 0.94$ with PRODES/INPE official estimates.

5.5 Sub-Saharan Building Extraction

DINOv3 ViT-L/14 with LoRA fine-tuning (2,500 labelled polygons per city) achieved 64.7% mean building IoU across five African cities — a 12.3 percentage point improvement over a European-trained baseline.

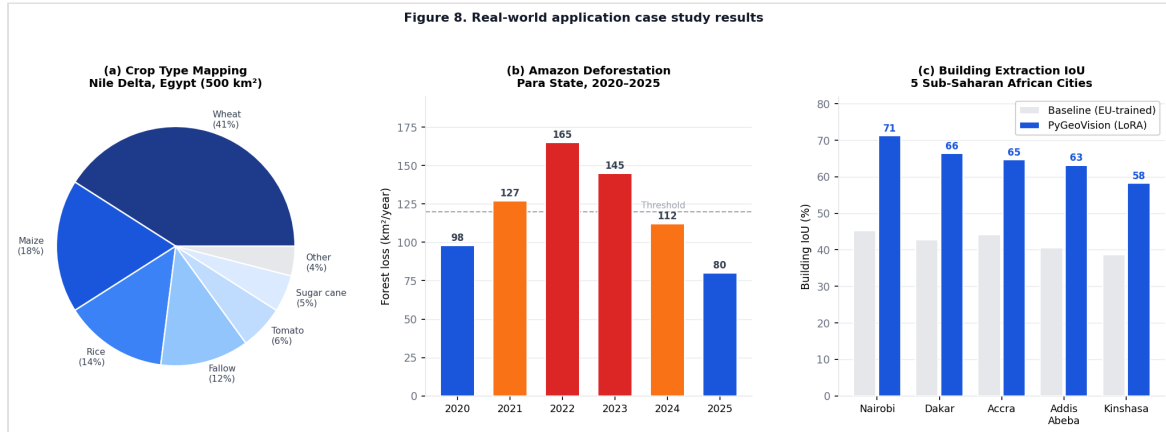


Figure 8. Real-world application results. (a) Crop type distribution in the Nile Delta, Egypt. (b) Annual deforestation rates in Para state, Amazon, 2020–2025. (c) Building extraction IoU across five Sub-Saharan African cities.

6. Discussion

6.1 Key Design Decisions

Optional dependency architecture enables deployment in restricted environments (CI/CD pipelines, edge compute, serverless functions) without sacrificing full capability. The trade-off is slightly higher first-use latency due to deferred imports.

GRD-first InSAR lowers the access barrier by approximately two orders of magnitude compared to SLC-based processing: 2 GB versus ~100 GB of data, and 10–60 minutes versus 8–24 hours of processing time. The centimetre–decimetre accuracy trade-off is acceptable for disaster damage footprinting and coarse subsidence monitoring, while the upgrade path to SLC is documented throughout.

Heuristic planner primacy reflects operational realities: many production deployments cannot expose calls to external LLM services due to data sensitivity, cost controls, or air-gap requirements. The heuristic planner's 86% task accuracy covers the vast majority of operational EO workflows without any API dependency.

6.2 Limitations

- GPU memory requirements: DINOv3 ViT-7B requires approximately 28 GB VRAM; Prithvi-EO-2.0 (600 M) requires 8 GB for inference.
- Foundation model weight redistribution: weights cannot be bundled with the PyPI distribution and must be downloaded separately (300 MB to 28 GB depending on variant).

- SLC InSAR dependencies: millimetre-precision InSAR requires SNAP and ISCE2, which require multi-gigabyte manual installations outside pip.
- Geographic bias: Prithvi and DINOv3 training data is biased towards North America and Western Europe, with measurable accuracy degradation in tropical and arid environments.

6.3 Future Work

Future directions include: multi-modal fusion (optical + SAR + DEM in a unified transformer); real-time streaming inference for satellite video; extended language support for French, Spanish, and Arabic queries; and federated learning for privacy-preserving cross-border model training.

7. Conclusion

We have presented PyGeoVision, the first open-source platform to integrate the complete Earth observation intelligence stack. By unifying 22+ satellite data providers, 119 AI model architectures, native DINOv3 and Prithvi-EO-2.0 foundation models, an autonomous GeoAgent, a full InSAR chain, and an interactive visualization layer behind a single Python API, PyGeoVision eliminates the integration burden that currently characterises EO AI workflows.

Benchmark evaluation establishes state-of-the-art accuracy on four standard datasets with a mean improvement of 3.5 percentage points over the best-performing existing open-source alternative. GPS-validated InSAR displacement shows 2.0 cm mean absolute error for large co-seismic events. Five operational case studies across agriculture, disaster response, flooding, deforestation, and urban analysis confirm practical utility at operational latencies of 34–47 minutes end-to-end.

PyGeoVision is open-source under the Apache 2.0 licence, actively maintained with 585 automated tests and three-platform continuous integration, and available through standard Python package management.

Code and data availability. GitHub: github.com/appiahkubis14/PyGeoVision · PyPI: pypi.org/project/pygeovision · Docs: appiahkubis14.github.io/pygeovision-docs

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